

Applied Mathematics for Chemists

(화학수학)

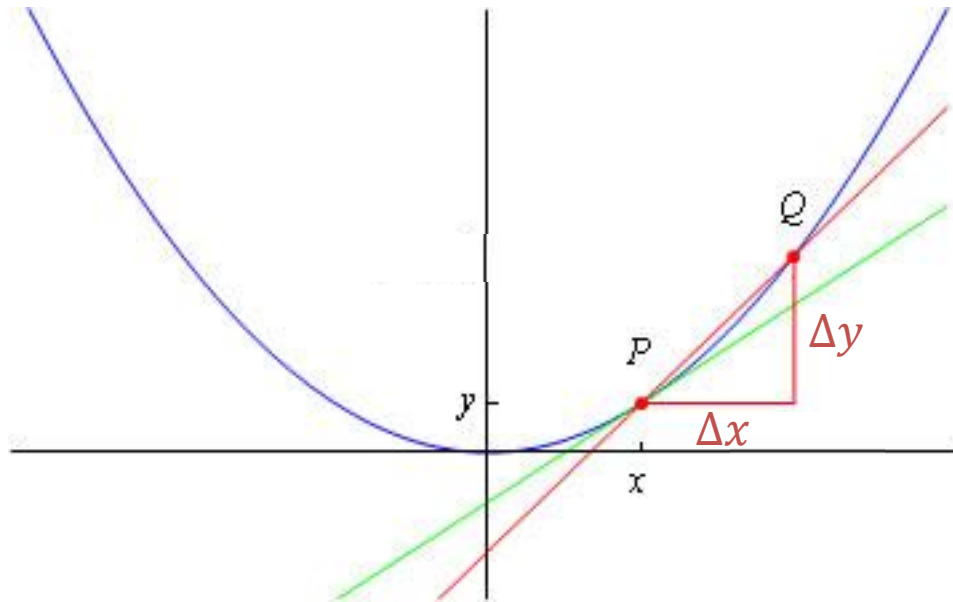
Lecture 4

Multivariate Differential Calculus



Differentiation

“The slope of a function”

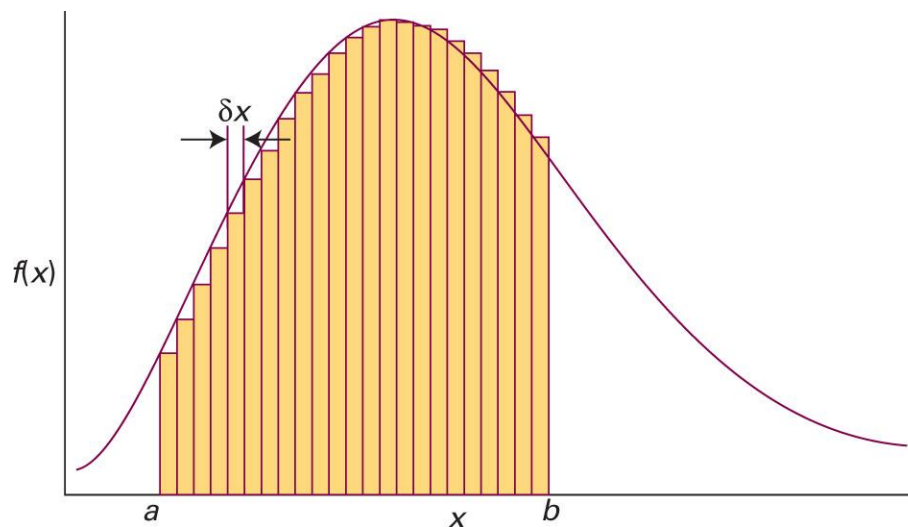


$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

(Image: http://wwwf.imperial.ac.uk/metric/metric_public/differentiation/index.html)

Integration

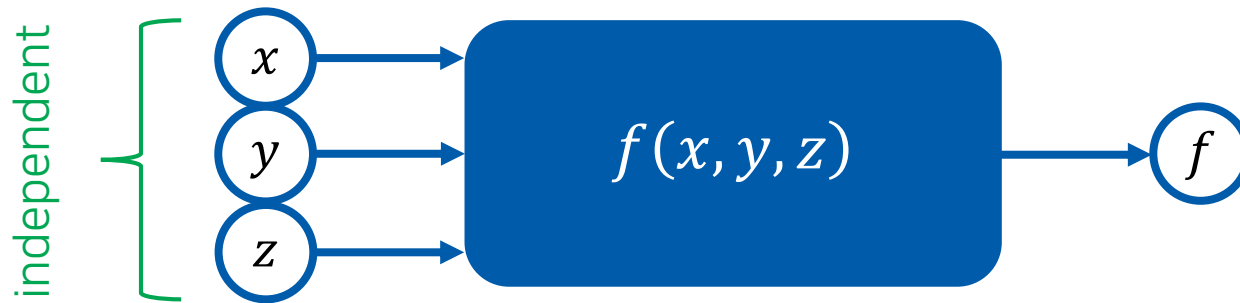
“The areas under curves”



$$\lim_{\delta x \rightarrow 0} \sum_i f(x_i) \delta x = \int_a^b f(x) dx$$

Function of Several Independent Variables

(다변수함수)



Example: $f(x, y, z) = x^2 + xy + 3z$

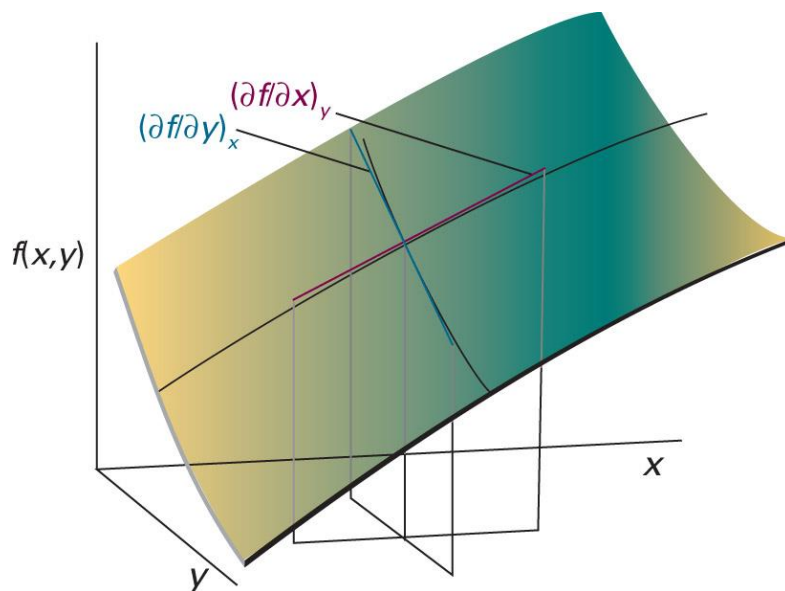
$$f(1, 1, 1) = 1 + 1 + 3 = 5$$

$$f(-1, 1, 0) = 1 - 1 + 0 = 0$$

⋮

Partial Derivatives (편미분)

“the slope of the function with respect to one of the variables,
all the other variables being held constant”



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Fundamental Equation of Differential Calculus

(미분학의 기본 방정식)

- n independent variables: $x_1, x_2, \dots, x_n$
- dependent variable: y

The differential of y is

$$dy = \sum_{i=1}^n \left(\frac{\partial y}{\partial x_i} \right)_{x'} dx_i$$

where x' means keeping all of the variables except for x_i fixed

Today's Class

- Manipulating total differentials
 - Change of variables
 - Variable-change identity
 - Reciprocal identity, cycle rule, chain rule
 - Euler reciprocity relation
- Exact and inexact differentials
 - Test for exactness
 - Integrating factor



Change of Variables

We can use different sets of independent variables:

- $U = U(T, V, n)$

$$dU = \left(\frac{\partial U}{\partial T} \right)_{V,n} dT + \left(\frac{\partial U}{\partial V} \right)_{T,n} dV + \left(\frac{\partial U}{\partial n} \right)_{T,V} dn$$

- $U = U(T, P, n)$

$$dU = \left(\frac{\partial U}{\partial T} \right)_{P,n} dT + \left(\frac{\partial U}{\partial P} \right)_{T,n} dP + \left(\frac{\partial U}{\partial n} \right)_{T,P} dn$$

Be Careful

For most systems,

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} \neq \left(\frac{\partial U}{\partial T}\right)_{P,n}$$

Fixed variables (subscripts) are important!

Example 8.3

Express the function $z = z(x, y) = ax^2 + bxy + cy^2$ in terms of x and u , where $u = xy$. Find the two partial derivatives $(\partial z / \partial x)_y$ and $(\partial z / \partial x)_u$.

As $y = u/x$, substitution leads to a change of variables:

$$z(x, u) = ax^2 + bx \left(\frac{u}{x} \right) + c \left(\frac{u}{x} \right)^2 = ax^2 + bu + \frac{cu^2}{x^2}$$

Now, the partial derivatives are

$$\begin{aligned} \left(\frac{\partial z}{\partial x} \right)_y &= 2ax + by \\ \left(\frac{\partial z}{\partial x} \right)_u &= 2ax - \frac{2cu^2}{x^3} = 2ax - \frac{2cy^2}{x} \end{aligned}$$

different!

Relating Different Sets of Subscripts

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + ?$$

Start from one set of independent variables:

$$U = U(T, P, n)$$

$$dU = \boxed{\left(\frac{\partial U}{\partial T}\right)_{P,n}} dT + \left(\frac{\partial U}{\partial P}\right)_{T,n} dP + \left(\frac{\partial U}{\partial n}\right)_{T,P} dn$$

Relating Different Sets of Subscripts

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + ?$$

$$dU = \left(\frac{\partial U}{\partial T}\right)_{P,n} dT + \left(\frac{\partial U}{\partial P}\right)_{T,n} dP + \left(\frac{\partial U}{\partial n}\right)_{T,P} dn$$

Divide both sides by dT :

$$\frac{dU}{dT} = \left(\frac{\partial U}{\partial T}\right)_{P,n} \frac{\cancel{dT}}{1} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \frac{dP}{dT} + \left(\frac{\partial U}{\partial n}\right)_{T,P} \frac{dn}{dT}$$

Relating Different Sets of Subscripts

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + ?$$

$$\frac{dU}{dT} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \frac{dP}{dT} + \left(\frac{\partial U}{\partial n}\right)_{T,P} \frac{dn}{dT}$$

Apply the constraints:

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \left(\frac{\partial P}{\partial T}\right)_{V,n} + \left(\frac{\partial U}{\partial n}\right)_{T,P} \cancel{\left(\frac{\partial n}{\partial T}\right)_{V,n}}$$

0

Variable-Change Identity

(변수-변화 항등식)

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \left(\frac{\partial P}{\partial T}\right)_{V,n}$$

Example 8.4

Apply the foregoing method to the function in Example 8.3 and find the relation between $(\partial z / \partial x)_u$ and $(\partial z / \partial x)_y$.

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + ?$$

For $z = z(x, y)$,

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy$$

$$\frac{dz}{dx} = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \frac{dy}{dx}$$

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_u$$

Example 8.4

Apply the foregoing method to the function in Example 8.3 and find the relation between $(\partial z/\partial x)_u$ and $(\partial z/\partial x)_y$.

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_u$$

The function in Example 8.3 is $z = z(x, y) = ax^2 + bxy + cy^2$ and $u = xy$.

Hence,

$$\left(\frac{\partial z}{\partial y}\right)_x = bx + 2cy$$

$$\left(\frac{\partial y}{\partial x}\right)_u = \left(\frac{\partial}{\partial x} \left[\frac{u}{x}\right]\right)_u = -\frac{u}{x^2} = -\frac{y}{x}$$

Example 8.4

Apply the foregoing method to the function in Example 8.3 and find the relation between $(\partial z/\partial x)_u$ and $(\partial z/\partial x)_y$.

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_u$$

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + (bx + 2cy) \left(-\frac{y}{x}\right)$$

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y - by - \frac{2cy^2}{x}$$

Reciprocal Identity

(역 항등식)

$$\left(\frac{\partial y}{\partial x}\right)_{z,u} = \frac{1}{(\partial x / \partial y)_{z,u}}$$

Example 8.5

Show that

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = \frac{1}{(\partial V / \partial P)_{n,T}}$$

for an ideal gas.

Since $P = nRT/V$,

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = -\frac{nRT}{V^2}$$

Since $V = nRT/P$,

$$\left(\frac{\partial V}{\partial P}\right)_{n,T} = -\frac{nRT}{P^2} = -nRT \left(\frac{V}{nRT}\right)^2 = -\frac{V^2}{nRT}$$

Second Partial Derivatives

(2계 편도함수)

For $z = z(x, y)$,

$$\left(\frac{\partial^2 z}{\partial x^2}\right)_y = \left[\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x}\right)_y\right]_y$$

$$\frac{\partial^2 z}{\partial y \partial x} = \left[\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x}\right)_y\right]_x$$

(Mixed second partial derivative; 혼합 2계 편도함수)

Euler Reciprocity Relation

(오일러 역관계식)

For $z = z(x, y)$,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Example 8.6

Show that

$$\left(\frac{\partial^2 P}{\partial V \partial T}\right)_n = \left(\frac{\partial^2 P}{\partial T \partial V}\right)_n$$

for an ideal gas.

Since $P = nRT/V$,

$$\left(\frac{\partial^2 P}{\partial V \partial T}\right)_n = \left[\frac{\partial}{\partial V} \left(\frac{nR}{V}\right)\right]_{n,T} = -\frac{nR}{V^2}$$

and

$$\left(\frac{\partial^2 P}{\partial T \partial V}\right)_n = \left[\frac{\partial}{\partial T} \left(-\frac{nRT}{V^2}\right)\right]_{n,V} = -\frac{nR}{V^2}$$

Thermodynamic Relation

$$dU = TdS - PdV + \mu dn$$

For $U = U(S, V, n)$,

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V,n} dS + \left(\frac{\partial U}{\partial V}\right)_{S,n} dV + \left(\frac{\partial U}{\partial n}\right)_{S,V} dn$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,n}, \quad P = -\left(\frac{\partial U}{\partial V}\right)_{S,n}, \quad \mu = \left(\frac{\partial U}{\partial n}\right)_{S,V}$$

Maxwell Relations

$$T = \left(\frac{\partial U}{\partial S} \right)_{V,n}, \quad P = - \left(\frac{\partial U}{\partial V} \right)_{S,n}, \quad \mu = \left(\frac{\partial U}{\partial n} \right)_{S,V}$$

Since $(\partial^2 U / \partial V \partial S)_n = (\partial^2 U / \partial S \partial V)_n$,

$$\left(\frac{\partial T}{\partial V} \right)_{S,n} = - \left(\frac{\partial P}{\partial S} \right)_{V,n}$$

Relation between apparently unrelated quantities

Cycle Rule

(순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

Start from $y = y(x, z)$;

$$dy = \left(\frac{\partial y}{\partial x}\right)_z dx + \left(\frac{\partial y}{\partial z}\right)_x dz$$

$$\frac{dy}{dx} = \left(\frac{\partial y}{\partial x}\right)_z \frac{dx}{dx} + \left(\frac{\partial y}{\partial z}\right)_x \frac{dz}{dx}$$

1

Cycle Rule

(순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

$$\frac{dy}{dx} = \left(\frac{\partial y}{\partial x}\right)_z + \left(\frac{\partial y}{\partial z}\right)_x \frac{dz}{dx}$$

Fix y ;

$$\left(\frac{\cancel{\partial y}}{\partial x}\right)_y = \left(\frac{\partial y}{\partial x}\right)_z + \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y$$

0

Cycle Rule

(순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

$$0 = \left(\frac{\partial y}{\partial x}\right)_z + \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y$$

$$-1 = \frac{\left(\frac{\partial y}{\partial x}\right)_z}{\left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y} = \left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x$$

Example 8.8

For the function $y = z \ln x$, show that the cycle rule is valid.

Since $x = e^{y/z}$ and $z = y/(\ln x)$,

$$\left(\frac{\partial y}{\partial x}\right)_z = \frac{z}{x}, \quad \left(\frac{\partial x}{\partial z}\right)_y = -\frac{y}{z^2} e^{y/z}, \quad \left(\frac{\partial z}{\partial y}\right)_x = \frac{1}{\ln x}$$

Hence,

$$\begin{aligned} \left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x &= \frac{z}{x} \times \left(-\frac{y}{z^2} e^{y/z}\right) \times \frac{1}{\ln x} \\ &= \frac{z}{x} \times \left(-\frac{y}{z^2} x\right) \times \frac{z}{y} = -1 \end{aligned}$$

Chain Rule

(연쇄 법칙)

For $z = z(u, x, y)$ and $x = x(u, v, y)$,

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = \left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v}$$

Note:

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v} \left(\frac{\partial y}{\partial z}\right)_{u,v} = 1$$

Check the subscripts!

Example 8.9

Show that if $z = ax^2 + bvx$ and $x = uy$, then the chain rule is valid.

Recall the chain rule:

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = \left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v}$$

Since $z = z(y, u, v) = a(uy)^2 + bv(uy) = au^2y^2 + buvy$,

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = 2au^2y + buv$$

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} = 2ax + bv, \quad \left(\frac{\partial x}{\partial y}\right)_{u,v} = u$$

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v} = (2ax + bv) \times u = 2au^2y + buv$$

same!

Pfaffian Form: General Differential Form

(파피안형)

$$du = M(x, y)dx + N(x, y)dy$$

- Exact differential (완전 미분):

$$M(x, y) = \left(\frac{\partial u}{\partial x} \right)_y, \quad N(x, y) = \left(\frac{\partial u}{\partial y} \right)_x$$

- Inexact differential (불완전 미분): otherwise

Test of Exactness

$$du = M(x, y)dx + N(x, y)dy$$

Using the Euler reciprocity relation,

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

if the differential is exact,

$$\left(\frac{\partial N}{\partial x} \right)_y = \left(\frac{\partial M}{\partial y} \right)_x$$

Example 8.11

Show that the following differential is exact:

$$dz = \underbrace{\left(2xy + \frac{9x^2}{y}\right)}_{M(x,y)} dx + \underbrace{\left(x^2 - \frac{3x^3}{y^2}\right)}_{N(x,y)} dy$$

Using the previous criterion,

$$\left(\frac{\partial M}{\partial y}\right)_x = \left[\frac{\partial}{\partial y}\left(2xy + \frac{9x^2}{y}\right)\right]_x = 2x - \frac{9x^2}{y^2}$$

$$\left(\frac{\partial N}{\partial x}\right)_y = \left[\frac{\partial}{\partial x}\left(x^2 - \frac{3x^3}{y^2}\right)\right]_y = 2x - \frac{9x^2}{y^2}$$

Since the two terms are equal, the differential is exact.

Test of Exactness: Three Variables

$$du = M(x, y, z)dx + N(x, y, z)dy + P(x, y, z)dz$$

If the differential is exact,

$$\left[\begin{array}{l} \left(\frac{\partial M}{\partial y} \right)_{x,z} = \left(\frac{\partial N}{\partial x} \right)_{y,z} \\ \left(\frac{\partial N}{\partial z} \right)_{x,y} = \left(\frac{\partial P}{\partial y} \right)_{x,z} \\ \left(\frac{\partial M}{\partial z} \right)_{x,y} = \left(\frac{\partial P}{\partial x} \right)_{y,z} \end{array} \right.$$

Inexact Differentials in Thermodynamics

- δq = the amount of heat transferred to the system
- δw = the amount of work done on the system

For a reversible process, $\delta w_{\text{rev}} = -PdV$

Exactness test: for $\delta w_{\text{rev}} = MdT + NdV$,

$$\left(\frac{\partial M}{\partial V}\right)_T = 0, \quad \left(\frac{\partial N}{\partial T}\right)_V = -\frac{nR}{V} \neq 0$$

The differential is inexact

First Law of Thermodynamics

$$dU = \delta q + \delta w$$

dU is an exact differential
even though δq and δw are not

Integrating Factor

(적분인자)

inexact differential $\rightarrow$ exact differential
by multiplying a function (integrating factor)

No general method to find an integrating factor

Example 8.13

Show that the differential

$$du = Mdx + Ndy = (2ax^2 + bxy)dx + (bx^2 + 2cxy)dy$$

is inexact, but that $1/x$ is an integrating factor, so that du/x is exact.

First, test the exactness:

$$\left(\frac{\partial M}{\partial y}\right)_x = \left[\frac{\partial}{\partial y}(2ax^2 + bxy)\right]_x = bx$$

$$\left(\frac{\partial N}{\partial x}\right)_y = \left[\frac{\partial}{\partial x}(bx^2 + 2cxy)\right]_y = 2bx + 2cx$$

so the differential is inexact.

Example 8.13

Show that the differential

$$du = Mdx + Ndy = (2ax^2 + bxy)dx + (bx^2 + 2cxy)dy$$

is inexact, but that $1/x$ is an integrating factor, so that du/x is exact.

Now consider du/x :

$$du/x = (2ax + by)dx + (bx + 2cy)dy = Pdx + Qdy$$

so

$$\left(\frac{\partial P}{\partial y}\right)_x = \left[\frac{\partial}{\partial y}(2ax + by)\right]_x = b$$

$$\left(\frac{\partial Q}{\partial x}\right)_y = \left[\frac{\partial}{\partial x}(bx + 2cy)\right]_y = b$$



Hence, the differential is exact.

Second Law of Thermodynamics

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

dS is an exact differential
even though δq_{rev} is not
(integrating factor: $1/T$)